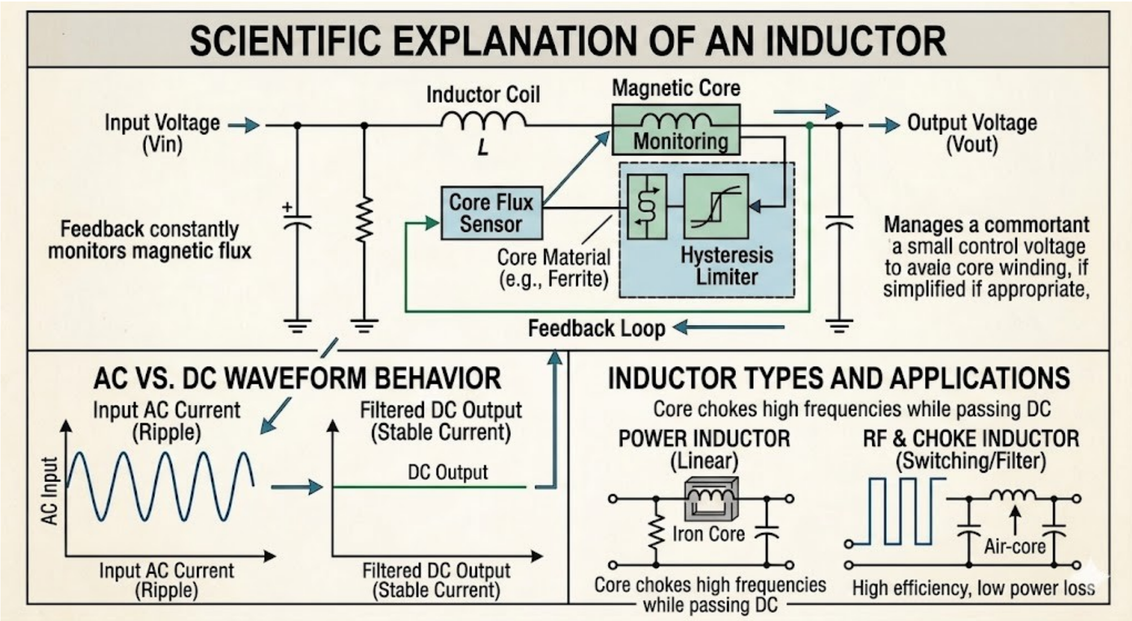
Physics Guide: COMPREHENSİVE TECHNİCAL REPORT: Inductor

Classification: TECHNICAL GUIDE/ EDUCATIONAL

Xumar Bağıyeva, Aysel İsmayılova, Rəşadə Zülfüqarlı, Yasin Quluzade,Ahmed Abdullayev

Verified by: Physics Teacher Azerbaijan Telman Askeraliyev (Fizika Muellimi, Azerbaijan, Baku)



1. Top Panel: MAIN CIRCUIT & WORKING PRINCIPLE

This section explains how an inductor functions within a dynamic circuit, utilizing a feedback mechanism to monitor magnetic flux.

- **SCIENTIFIC EXPLANATION OF AN INDUCTOR:** An inductor is a passive electronic component that stores energy in a magnetic field when electric current flows through it. Its core property is inertia against changes in current.
- **Input Voltage (V_{in}) & Capacitor:** The unregulated input enters the circuit, where a parallel capacitor helps suppress initial high-frequency voltage spikes.
- **Inductor Coil (L):** As current flows through the coil, it creates a magnetic field. According to **Lenz's Law**, if the input current tries to change rapidly, the coil induces a Back EMF (electromotive force) to oppose that change.

- **Magnetic Core & Flux Monitoring:** The inductor wraps around a core material (like Ferrite or Iron) to concentrate the magnetic field. The system constantly monitors the magnetic flux density to prevent **core saturation** (the point where the core cannot hold more magnetic lines of force).
- **Hysteresis Limiter & Core Flux Sensor:** This control loop ensures the magnetic field changes smoothly, stabilizing the output voltage (V_{out}) and protecting the system against rapid thermal or magnetic overloads.

---

2. Bottom-Left Panel: AC vs. DC WAVEFORM BEHAVIOR

This panel visualizes how the inductor fundamentally acts as a low-pass filter for signals.

- **Input AC Current (Ripple):** The raw input current contains heavy fluctuations, noise, and high-frequency switching ripples.
- **The Inductor's Action:** High frequencies meet high resistance from the inductor, because its opposition to current—called **Inductive Reactance (X_L)**—increases as frequency goes up ($X_L = 2\pi f L$).
- **Filtered DC Output (Stable Current):** Because steady DC has a frequency of zero ($f = 0$), the inductor treats it like a simple wire (short circuit), letting the clean, flat DC pass through completely unhindered.

---

3. Bottom-Right Panel: INDUCTOR TYPES & APPLICATIONS

This section classifies inductors based on their physical core construction and operational roles.

- **Power Inductor (Iron Core):** Uses a heavy iron core to handle massive currents and large amounts of energy. These are commonly used in linear power grids and heavy smoothing filters because iron can sustain high magnetic saturation limits.
- **RF & Choke Inductor (Air-Core / Switching):** Uses an air core or ferrite core designed for high-frequency switching operations (like Buck or Boost converters). It quickly builds and collapses its magnetic field with minimal power loss, effectively "choking" high-frequency RF noise while maintaining top efficiency.

Mövzunu tam dolğun və peşəkar səviyyəyə çatdırmaq üçün bura əlavə edə biləcəyin, təqdimatda və ya layihə sənədində (report) mütləq istifadə olunan **elmi tənliklər**, **əsas parametrlər** və **praktiki tətbiq sahələrini** təqdim edirəm.

Bunları birbaşa kopyalayıb mətninə yapışdıra bilərsən:

---

4. CORE MATHEMATICAL FORMULAS (Elmi Tənliklər)

An inductor's behavior is fully governed by precise electromagnetic equations. You can use these to show the mathematical background:

- **Voltage-Current Relationship (Ohm's Law for Inductors):**

The voltage across an inductor is directly proportional to how fast the current changes through it:

$$V = L \frac{di}{dt}$$

(Where V is induced voltage, L is inductance in Henrys, and $\frac{di}{dt}$ is the rate of change of current over time).

- **Inductive Reactance (Opposition to AC):**

Unlike a standard resistor, an inductor's resistance changes with the frequency (f) of the signal:

$$X_L = 2\pi f L$$

(This proves mathematically why high frequencies are blocked/choked, while DC ($f=0$) passes safely with zero reactance).

- **Energy Storage Capacity:**

The actual kinetic energy stored inside the inductor's magnetic field is calculated using:

$$E = \frac{1}{2} L I^2$$

(Where E is energy in Joules, and I is the current flowing through the coil).

---

5. CRITICAL DESIGN PARAMETERS (Əsas Seçim Meyarları)

When selecting an inductor for a real-world circuit diagram, engineers must evaluate these specific parameters:

- **Self-Resonant Frequency (SRF):** The frequency at which the inductor's internal parasitic capacitance cancels out its inductance. Beyond this frequency, the inductor stops acting like an inductor and begins acting like a capacitor.
- **Saturation Current (I_{sat}):** The maximum current level where the magnetic core material can no longer hold any more magnetic flux lines. Exceeding this causes the inductance value (L) to drop violently.
- **DC Resistance (DCR):** The pure resistance of the copper wire wrapped inside the coil. A lower DCR means less power is wasted as heat (I^2R loss), which maximizes efficiency.

---

6. REAL-WORLD INDUSTRIAL APPLICATIONS (Praktiki Tətbiq Sahələri)

- **Switch-Mode Power Supplies (SMPS):** Used in Buck and Boost converters to temporarily store energy when the high-frequency transistor switches on and off.
- **EMI/RFI Filtering:** Placed at the power input lines of computers and sensitive medical equipment to block radio frequency interference and high-frequency noise coming from the main grid.
- **Transformers:** Two inductors placed close to each other, sharing magnetic flux lines, allowing voltage to step-up or step-down through mutual induction.